

Section : Physics

Q.1 Identify the correct statements.

- (A) The energy (E) and momentum (p) of a photon are related by $p = E/c$.
- (B) The electrons and photons of the same wavelength, possess the same momentum.
- (C) One photon having energy more than work function can emit one photo electron.
- (D) Photons do not carry momentum.

Choose the **correct** answer from the options given below:

- 1. (A), (C) and (D) only
- 2. (B), (C) and (D) only
- 3. (C) and (D) only
- 4. (A), (B) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.2 Which of the following are true for optical instruments?

- (A) A simple microscope is a converging lens of large focal length.
- (B) A telescope is used to provide angular magnification of distant objects.
- (C) The objective lens of a telescope has a smaller focal length than the eyepiece.
- (D) The magnifying power of a telescope is the ratio of the focal length of the objective lens to the focal length of the eyepiece.

(E) A simple microscope is a converging lens of small focal length.

Choose the **correct** answer from the options given below:

- 1. (A), (B) and (C) only
- 2. (B), (C) and (D) only
- 3. (B), (D) and (E) only
- 4. (A), (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.3 An earthed conductor is placed near another positively charged conductor. The net charge acquired by the earthed conductor will be:

- 1. Positive only
- 2. Negative only
- 3. Zero
- 4. Both positive and negative on the far and near sides, respectively

Options 1. 1

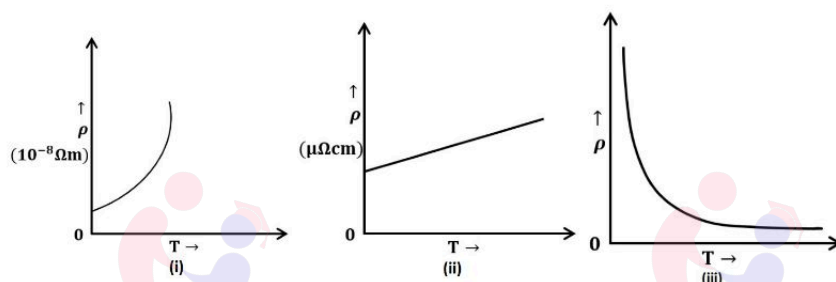
2. 2

3. 3

4. 4

Q.4

Identify the correct statement(s) from the following Resistivity (ρ) versus Temperature (T) graphs:



(A) Graph (i), shows ρ versus T variation for Copper

(B) Graph (ii), shows ρ versus T variation for Silicon

(C) Graph (iii), shows ρ versus T variation for semiconductor

(D) Graph (iii), shows ρ versus T variation for Nichrome

Choose the **correct** answer from the options given below:

- 1. (A) only
- 2. (A) and (C) only
- 3. (B) and (C) only
- 4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.5 The plates of a parallel plate capacitor are separated by a distance d with air as medium between the plates. In order to increase its capacitance by 60%, a slab of dielectric constant 5 having the same area as that of plates is introduced between the plates. The thickness of the dielectric slab is:

1. $\frac{5}{8}d$
2. $\frac{8}{23}d$
3. $\frac{6}{19}d$
4. $\frac{15}{32}d$

Options

1. 1
2. 2
3. 3
4. 4

Q.6 In the hydrogen atom, an electron revolves around the nucleus in a circular orbit of radius 5.11×10^{-11} m with a frequency of 6.8×10^{15} Hz. The magnetic field at the centre of the orbit is:

1. 6.7 T
2. 13.4 T
3. 20.1 T
4. 26.8 T

Options

1. 1
2. 2
3. 3
4. 4

Q.7 In a series LR circuit, the inductive reactance is equal to the resistance R of the circuit.

An emf $E = E_0 \cos \omega t$ is applied to the circuit. The average power dissipated in one cycle is:

1. $\frac{E_0^2}{R}$
2. $\frac{E_0^2}{4R}$
3. $\frac{E_0^2}{2R}$
4. $\frac{E_0^2}{8R}$

Options

1. 1
2. 2
3. 3
4. 4

Q.8 In the formation of a p-n junction diode, arrange the following processes in sequence.

- (A) Diffusion of charge carriers
- (B) Drift of charge carriers
- (C) Doping
- (D) Generation of electric field at the junction

Choose the correct answer from the options given below:

1. (C), (A), (B), (D)
2. (D), (C), (A), (B)
3. (C), (A), (D), (B)
4. (C), (D), (A), (B)

Options

1. 1
2. 2
3. 3
4. 4

Q.9 A 15 cm wire carrying a current of 2 A is placed inside a current-carrying solenoid, perpendicular to its axis.

The magnetic force acting on the wire is 20×10^{-2} N.

The magnetic field inside the solenoid is:

1. 1.5×10^{-4} T

2. 0.67 T

3. $3.0 \times 10^{-4} \text{ T}$

4. $3.33 \times 10^{-3} \text{ T}$

- Options
- 1
 - 2
 - 3
 - 4

Q.10 The focal length of a convex lens in air and in a liquid medium are 20 cm and 80 cm, respectively. If the absolute refractive index of lens is 1.5, then the refractive index of liquid is:

1. $3/2$
2. $4/3$
3. $8/9$
4. $9/8$

- Options
- 1
 - 2
 - 3
 - 4

Q.11 The nucleus ${}^{23}_{10}\text{Ne}$ decays by β^- emission. The maximum kinetic energy of the emitted electrons is:
Given that:

$$m({}^{23}_{10}\text{Ne}) = 22.994466 \text{ u}, m({}^{23}_{11}\text{Na}) = 22.989770 \text{ u} \text{ and } 1\text{u} = 931 \text{ MeV}/c^2$$

1. 3.472 MeV
2. 4.732 MeV
3. 2.375 MeV
4. 4.372 MeV

- Options
- 1
 - 2
 - 3
 - 4

Q.12 Match List-I with List-II

List-I	List-II
Materials	Band Gap (eV)
(A) Germanium	(I) 1.42 eV
(B) Gallium Arsenide	(II) $> 3 \text{ eV}$
(C) Insulator	(III) 1.1 eV
(D) Silicon	(IV) 0.7 eV

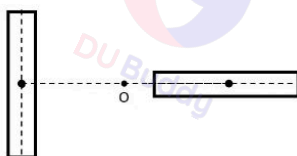
Choose the correct answer from the options given below:

1. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
2. (A) - (IV), (B) - (I), (C) - (III), (D) - (II)
3. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
4. (A) - (IV), (B) - (II), (C) - (I), (D) - (III)

- Options
- 1
 - 2
 - 3
 - 4

Q.13

Two identical short magnetic dipoles of magnetic moment 2 Am^2 are placed with their axis mutually perpendicular and centers 2 m apart. The magnitude of the resultant magnetic field at the midpoint (O) between the dipoles is:



1. $2\sqrt{5} \times 10^{-7} \text{ T}$
2. $4\sqrt{5} \times 10^{-7} \text{ T}$
3. $2\sqrt{5} \times 10^{-6} \text{ T}$
4. $4\sqrt{5} \times 10^{-6} \text{ T}$

- Options
- 1
 - 2
 - 3

Q.14 In Young's double slit experiment, two slits are 2 mm apart and the screen is 2 m away from the slits.

A light source having two wavelengths 4000 Å and 6000 Å is used.

The minimum distance from the centre of the screen where the bright fringes produced by both wavelengths coincide is:

1. 1.2×10^{-4} m
2. 1.2×10^{-3} m
3. 1.8×10^{-4} m
4. 2.4×10^{-3} m

Options

1. 1
2. 2
3. 3
4. 4

Q.15 In an ac circuit, a coil of self-inductance L is connected in series with an electric bulb of resistance R . If the frequency of ac supply increases, then

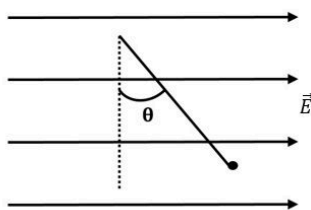
1. Impedance of the circuit decreases
2. Effective current in the circuit increases
3. Brightness of the bulb decreases
4. Phase difference between the current and the voltage decreases

Options

1. 1
2. 2
3. 3
4. 4

Q.16

A pendulum of length ' l ' having a bob of mass ' m ' carrying charge ' q ' is suspended from a stationary rigid support in a region of uniform electric field ' E ' as shown in the figure. If the system is in equilibrium, the nature and magnitude of ' q ' will be:



1. Positive, $\frac{mg}{E}$
2. Positive, $\left(\frac{mg}{E}\right) \tan \theta$
3. Positive, $\frac{mg}{E \tan \theta}$
4. Negative, $\left(\frac{mg}{E}\right) \tan \theta$

Options

1. 1
2. 2
3. 3
4. 4

Q.17 The kinetic energy of an electron in a given orbit in a hydrogen atom is:

(symbols have their usual meaning)

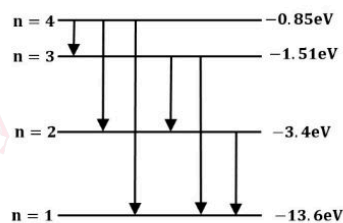
1. $\frac{e^2}{4\pi\epsilon_0 r}$
2. $\frac{e^3}{8\pi\epsilon_0 r}$
3. $\frac{e^2}{8\pi\epsilon_0 r}$
4. $\frac{e^2}{3\pi\epsilon_0 r}$

Options

1. 1
2. 2
3. 3
4. 4

Q.18

The figure shows the energy level diagram of the hydrogen atom. Out of the six given transitions, which transition results in the emission of a photon of wavelength $4870 \text{ \AA}$? ($h = 6.63 \times 10^{-34} \text{ Js}$)



1. $n = 4$ to $n = 2$
2. $n = 3$ to $n = 1$
3. $n = 4$ to $n = 1$
4. $n = 3$ to $n = 2$

Options

1. 1
2. 2
3. 3
4. 4

Q.19

In a Young's double slit experiment, the n^{th} bright fringe for a light of wavelength $5000 \text{ \AA}$ coincides with the $(n + 1)^{\text{th}}$ bright fringe for light of wavelength $4000 \text{ \AA}$.

The value of n is:

1. 3
2. 5
3. 2
4. 4

Options

1. 1
2. 2
3. 3
4. 4

Q.20

The self-inductance of an inductance coil having 100 turns is 20 mH.

The total flux through the coil for a current of 40 mA is:

1. $8 \times 10^{-3} \text{ Wb}$
2. $4 \times 10^{-3} \text{ Wb}$
3. $8 \times 10^{-4} \text{ Wb}$
4. $4 \times 10^{-4} \text{ Wb}$

Options

1. 1
2. 2
3. 3
4. 4

Q.21

A 16Ω wire is bent to form a square loop. A cell of 9 V is connected across one of its sides.

The potential difference between the diagonals of the square loop is:

1. 6V
2. 4.5 V
3. 8V
4. 9V

Options

1. 1
2. 2
3. 3
4. 4

Q.22

Match List-I with List-II for convex lens (symbols have their usual meanings)

List-I

List-II

Position of Object

Details of image

(A) at $2F$

(I) $m > 1$, virtual

(B) at F

(II) $m < 1$, real

(C) between lens and F

(III) $m = 1$, real

(D) between ∞ and $2F$

(IV) $m > 1$, real

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
2. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
3. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
4. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Options 1. 1

2. 2
3. 3
4. 4

Q.23 In the nuclear reaction below, a neutron hits a uranium nucleus.



The values of a and b are respectively:

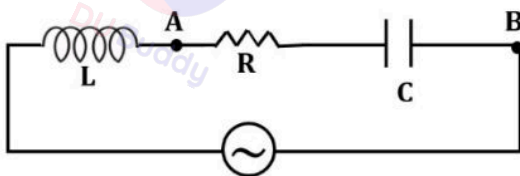
1. 141 and 36
2. 140 and 38
3. 139 and 38
4. 140 and 36

Options 1. 1

2. 2
3. 3
4. 4

Q.24

In the given circuit $L = \frac{20}{\pi}$ H, $R = 2 \text{ k}\Omega$ and $C = \frac{5}{\pi} \mu\text{F}$ are in series across a 220 V, 50 Hz ac source.



The peak voltage drop between A and B is:

1. 440 V
2. $220\sqrt{2}$ V
3. 480 V
4. 520 V

Options 1. 1

2. 2
3. 3
4. 4

Q.25 The magnitude of the electric and magnetic fields in an electromagnetic wave are related as:

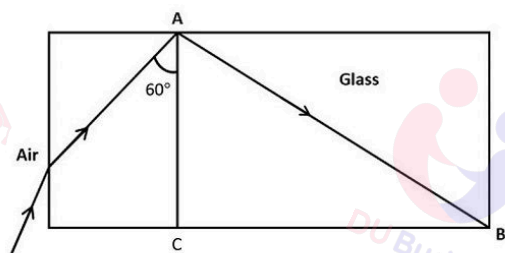
1. $E_0 = \frac{B_0}{c}$
2. $B_0 = \frac{E_0}{c}$
3. $E_0 B_0 = c$
4. $E_0 B_0 = \frac{1}{c}$

Options 1. 1

2. 2
3. 3
4. 4

Q.26

A light ray from air is incident on a glass slab ($\mu = 1.5$) as shown in the figure. It undergoes a total internal reflection and travels along AB. The thickness AC of the glass slab is 20 cm. The time taken by light to cross the glass slab after reflection is



1. 1.2 ns
2. 2 ns
3. 0.06 ns
4. 1.3 ns

Options 1. 1

2. 2
3. 3
4. 4

Q.27

The voltage and current in an a.c. circuit are:

$$V = 8 \sin(50\pi t - \pi/3) \text{ V}, \quad I = 6 \sin(50\pi t + \pi/3) \text{ A}.$$

The correct phase relationship between voltage and current is:

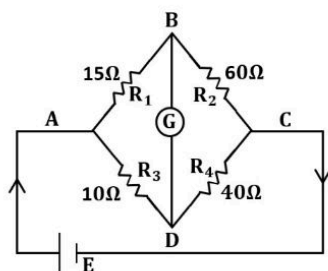
1. Voltage leads the current by 60°
2. Current leads the voltage by 60°
3. Current leads the voltage by 120°
4. Voltage leads the current by 120°

Options 1. 1

2. 2
3. 3
4. 4

Q.28

The figure shows a Wheatstone bridge circuit arrangement with a galvanometer G of resistance 5Ω to detect the presence of current. The ratio of heat dissipated across four resistors, R_1 , R_2 , R_3 , and R_4 respectively, is:



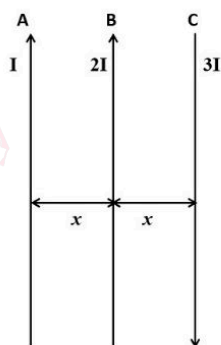
1. 2.25 : 9 : 1 : 4
2. 1 : 4 : 1.5 : 6
3. 1 : 2.25 : 4 : 9
4. 1.5 : 1 : 4 : 6

Options 1. 1

2. 2
3. 3
4. 4

Q.29

A, B, and C are parallel conductors of equal lengths carrying current I , $2I$ and $3I$ respectively, as shown in the figure. F_1 is the force exerted by B on A and F_2 is the force exerted by C on A. Choose the correct answer



1. $\vec{F}_1 = -\vec{F}_2$
2. $\vec{F}_2 = 2\vec{F}_1$
3. $\vec{F}_2 = \frac{3}{4}\vec{F}_1$
4. $\vec{F}_1 = -1.33\vec{F}_2$

Options

1. 1
2. 2
3. 3
4. 4

Q.30

A proton, a deuteron and an alpha particle have the same kinetic energy and move in a uniform magnetic field perpendicular to their direction of motion. The ratio of the radii of their circular paths is:

1. $1 : \sqrt{2} : 2$
2. $1 : 2 : \sqrt{2}$
3. $1 : \sqrt{2} : 4$
4. $1 : \sqrt{2} : 1$

Options

1. 1
2. 2
3. 3
4. 4

Q.31

Identify the correct statements

- (A) Electric current arises due to the continuous flow of electrons or ions.
- (B) A current flows through a conductor only when there is an electric field within the conductor.
- (C) The drift velocity of electrons is directly proportional to the applied electric field.
- (D) The average thermal velocity of the electrons in a conductor is zero.
- (E) In the absence of an electric field, the electrons in a conductor are at rest.

Choose the correct answer from the options given below:

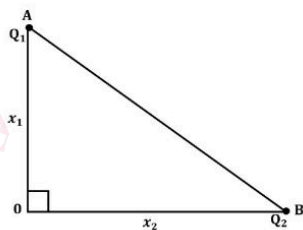
1. (A), (B), (C) and (D) only
2. (B), (C), (D) and (E) only
3. (A), (C), (D) and (E) only
4. (A), (B), (D), (E) only

Options

1. 1
2. 2
3. 3
4. 4

Q.32

AOB is a right angled triangle. Charges Q_1 and Q_2 are placed as shown. If the resultant electric field at O is perpendicular to the hypotenuse then $\frac{Q_1}{Q_2}$ is proportional to:



1. $\frac{x_1}{x_2}$
2. $\frac{x_2}{x_1}$
3. $\frac{x_1^2}{x_2^2}$
4. $\frac{x_2^2}{x_1^2}$

Options

1. 1
2. 2
3. 3
4. 4

Q.33 The ratio of magnetic fields at the centre of a current carrying a circular loop of radius 16 cm and at a point on its axis at a distance r from its centre is 125 : 64. The value of r is

1. 6 cm
2. 9 cm
3. 12 cm
4. 15 cm

Options

1. 1
2. 2
3. 3
4. 4

Q.34 Two solenoids of the same length and the same cross-sectional area have 400 turns and 250 turns, respectively. The ratio of self-inductance of the two solenoids, respectively, is

1. 64:25
2. 25:64
3. 8:5
4. 5:8

Options

1. 1
2. 2
3. 3
4. 4

Q.35 The resistance of a cylindrical conductor is:

1. inversely proportional to its length
2. directly proportional to the square of the radius
3. inversely proportional to the square of the radius
4. directly proportional to the square root of its length

Options

1. 1
2. 2
3. 3
4. 4

Q.36 Identify the correct properties of magnetic field lines.

- (A) Magnetic field lines are directed from the south pole to the north pole outside the magnet.
 - (B) The magnetic field lines never intersect each other.
 - (C) Magnetic lines always form a closed and continuous curve.
 - (D) If magnetic field lines are parallel, then the magnetic field must be uniform throughout.
- Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (B), (C) and (D) only
3. (A), (C) and (D) only
4. (B) and (C) only

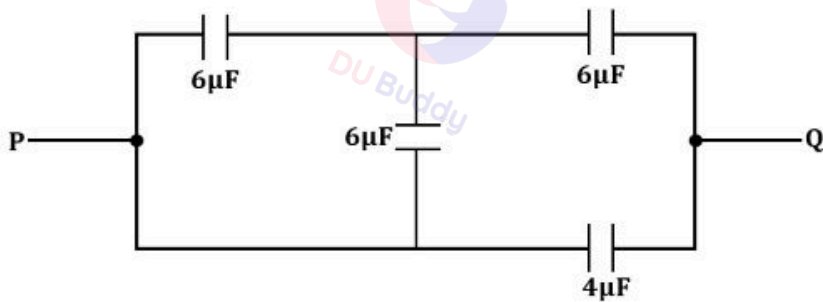
Options

1. 1

2. 2
3. 3
4. 4

Q.37

Four capacitors are connected in the circuit as shown.



The equivalent capacitance between P and Q will be:

1. $8\mu\text{F}$
2. $6\mu\text{F}$
3. $13\mu\text{F}$
4. $36/13\mu\text{F}$

Options

1. 1
2. 2
3. 3
4. 4

Q.38 216 charged droplets, each of radius r and charge q , are combined to form a big drop. The ratio of potential of the big drop to the small droplet will be:

1. 216:1
2. 49:1
3. 36:1
4. 7:1

Options

1. 1
2. 2
3. 3
4. 4

Q.39 Match List-I with List-II

List-I

List-II

Frequency Range

EM Waves

(A) $4 \times 10^{14} \text{ Hz} - 7.5 \times 10^{14} \text{ Hz}$ (I) Infrared rays

(A) $3 \times 10^{17} \text{ Hz} - 3 \times 10^{20} \text{ Hz}$ (II) Microwaves

(A) $3 \times 10^9 \text{ Hz} - 3 \times 10^{11} \text{ Hz}$ (III) Visible Light

(A) $3 \times 10^{11} \text{ Hz} - 4 \times 10^{14} \text{ Hz}$ (IV) X-rays

Choose the correct answer from the options given below:

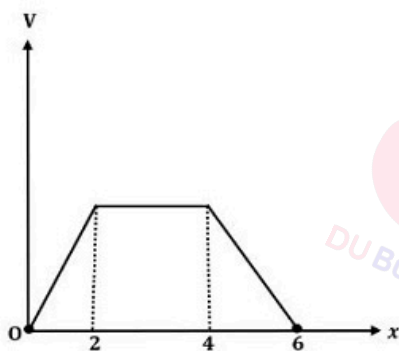
1. (A) - (III), (B) - (I), (C) - (II), (D) - (IV)
2. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
3. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
4. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

Options

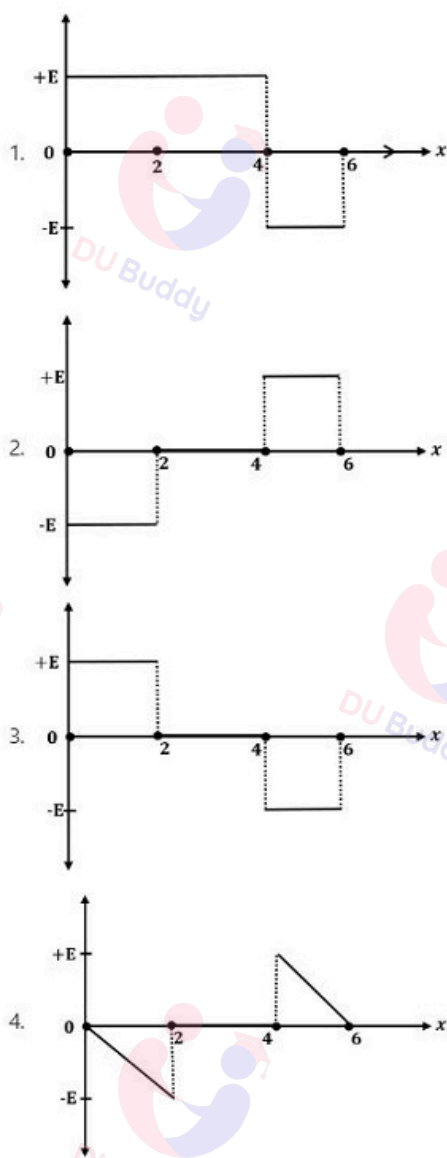
1. 1
2. 2
3. 3
4. 4

Q.40

The given figure shows the variation of electric potential V as a function of distance x .



The graph for the variation of electric field E as a function of distance x is:



Options 1. 1

2. 2

3. 3

4. 4

Q.41 Under forward-bias of a p-n diode

(A) p-side of the diode is connected to the positive terminal of the battery and n-side to the negative terminal

(B) p-side of the diode is connected to the negative terminal of the battery and n-side to the positive terminal

(C) The direction of the applied voltage (V) is opposite to the built-in potential drop

(D) The direction of the applied voltage (V) is the same as the built-in potential drop

(E) The magnitude of the diode forward current is usually in mA
Choose the *correct* answer from the options given below:

1. (A), (C) and (D) only
2. (A), (D) and (E) only
3. (A), (B), (C) and (D) only
4. (A), (C) and (E) only

Options

1. 1
2. 2
3. 3
4. 4

Q.42 When light travels from one medium to another medium of different refractive index, then the wave will have change in its:

1. Frequency, wavelength and velocity
2. Frequency and wavelength only
3. Frequency and velocity only
4. Wavelength and velocity only

Options

1. 1
2. 2
3. 3
4. 4

Q.43 If the work function of a target is 1.25 eV and the wavelength of incident radiation is $2460 \text{ \AA}$, the stopping potential required to stop the emitted electrons is:

Given: $h = 6.6 \times 10^{-34} \text{ Js}$

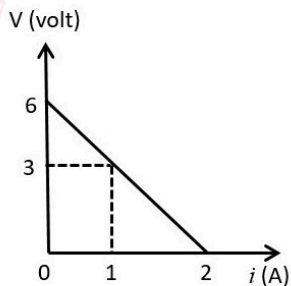
1. 1.75 V
2. 2.75 V
3. 3.75 V
4. 5.75 V

Options

1. 1
2. 2
3. 3
4. 4

Q.44

The given graph shows the variation of terminal potential difference V , across the combination of three identical cells in series to an external resistor versus the current i in the circuit. The internal resistance of each cell is:



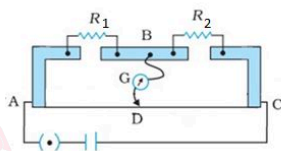
1. 0.5Ω
2. 1Ω
3. 1.5Ω
4. 0.75Ω

Options

1. 1
2. 2
3. 3
4. 4

Q.45

In the given setup of meter bridge, the null point D is obtained at a distance of 40 cm from end A. If a resistance of $10\ \Omega$ is connected in series with R_1 , the null points shift by 20 cm. The values of R_1 and R_2 respectively are



1. $8\ \Omega$ and $12\ \Omega$
2. $12\ \Omega$ and $8\ \Omega$
3. $6\ \Omega$ and $4\ \Omega$
4. $6\ \Omega$ and $9\ \Omega$

Options 1. 1

2. 2

3. 3

4. 4

Q.46 Arrange the following electromagnetic waves in the order of their decreasing frequency.

- (A) Radio wave
- (B) Gamma rays
- (C) Ultraviolet ray
- (D) Infrared wave
- (E) Visible light

Choose the correct answer from the options given below:

1. (B) > (C) > (E) > (D) > (A)
2. (B) > (C) > (D) > (E) > (A)
3. (C) > (B) > (E) > (A) > (D)
4. (C) > (B) > (E) > (D) > (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.47

The electric field in a region is given by $E = \frac{E_0 x}{b}$. Find the charge contained in the cubical volume bounded by the

$x = 0$, $x = a$, $y = 0$, $y = a$ and $z = 0$, $z = a$. Take $E_0 = 6 \times 10^3\ \text{N/C}$, $a = 1\ \text{cm}$ and $b = 2\ \text{cm}$

1. $2.2 \times 10^{-12}\ \text{C}$
2. $3.5 \times 10^{-12}\ \text{C}$
3. $7.4 \times 10^{-13}\ \text{C}$
4. $2.65 \times 10^{-12}\ \text{C}$

Options 1. 1

2. 2

3. 3

4. 4

Q.48 Match List-I with List-II

List-I

Law/Rule

List-II

Application

(A) Faraday's law

(I) Determines direction of force on a current carrying wire in a magnetic field

(B) Lenz's law

(II) Obeys principle of energy conservation

(C) Fleming's right-hand rule

(III) Determines direction of induced current

(D) Fleming's left-hand rule

(IV) Electromagnetic Induction

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
2. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)
3. (A) - (IV), (B) - (II), (C) - (I), (D) - (III)
4. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.49

Light of frequency $8.27 \times 10^{14}\ \text{Hz}$ is incident on a metal surface. Electrons with maximum speed $6.6 \times 10^5\ \text{m/s}$ are ejected.

The threshold frequency for photoemission of electrons will be (given $h = 6.6 \times 10^{-34}\ \text{Js}$, and mass of electron = $m_e = 9 \times 10^{-31}\ \text{kg}$):

1. $5.30 \times 10^{14} \text{ Hz}$

2. $4.74 \times 10^{14} \text{ Hz}$

3. $3.84 \times 10^{14} \text{ Hz}$

4. $5.94 \times 10^{14} \text{ Hz}$

Options

1. 1

2. 2

3. 3

4. 4

Q.50 The light from a point source placed in air is incident on a convex spherical glass surface of radius of curvature 30 cm. The refractive index of the glass is 1.5. The distance of light source from the glass surface is 120 cm. The distance of the image formed from the glass surface is

1. 60 cm

2. 120 cm

3. 180 cm

4. 240 cm

Options

1. 1

2. 2

3. 3

4. 4